EXPERIMENT 5: EFFECT OF ROUTE OF ADMINISTRATION ON DRUG EFFECT

40 STANDARD IN-COURSE MCQs

1. The major aim of Experiment 5 is to determine:

A. The lethal dose of pentobarbital in male rats B. The effect produced when the same drug is administered into the body via different routes C. The rate of renal drug clearance in healthy mice D. The minimum inhibitory concentration of barbiturates

Answer: B

2. The experimental animals designated in the protocol were:

A. Female mice B. Male rats C. Female guinea pigs D. Male rabbits

Answer: B

3. Why are male animals preferred in pharmacokinetic experiments comparing routes?

A. Males completely lack hepatic enzymes B. To avoid estrous cycle hormonal fluctuations that alter drug metabolism C. Males do not possess a righting reflex D. Female animals cannot absorb drugs intraperitoneally

Answer: B

4. The drug listed on Page 41 is:

A. Diazepam B. Pentobarbitone or Nembutal C. Morphine sulphate D. Chlorpromazine

Answer: B

5. Pentobarbitone belongs to which pharmacological class?

A. Opioid analgesics B. Benzodiazepines C. Barbiturates D. Phenothiazines

Answer: C

6. The primary molecular mechanism of pentobarbitone involves:

A. Blocking voltage-gated sodium channels B. Prolonging the duration of chloride channel opening at the GABA-A receptor C. Stimulating central alpha-2 adrenergic receptors D. Inhibiting acetylcholinesterase

Answer: B

7. What is the proprietary / trade name for pentobarbital sodium listed on Page 41?

A. Luminal B. Valium C. Nembutal D. Seconal

Answer: C

8. The first procedural step instructed in the Method is:

A. Shave the animal's abdomen B. Weigh the first animal C. Administer oral glucose D. Place the animal on a heated grid

Answer: B

9. The standardized dose administered across all routes was:

A. 25 mg/kg B. 50 mg/kg C. 75 mg/kg D. 100 mg/kg

Answer: D

10. The concentration of the drug solution used in the laboratory was:

A. 1 mg/mL B. 5 mg/mL C. 10 mg/mL D. 50 mg/mL

Answer: C

11. The formula used to calculate the volume of drug solution to inject is:

A. Dose (mg/kg) × Weight (kg) ÷ Concentration (mg/mL) B. Concentration (mg/mL) × Weight (kg) ÷ Dose (mg/kg) C. Dose (mg/kg) × Concentration (mg/mL) × Weight (kg) D. Weight (kg) ÷ Dose (mg/kg) × Concentration (mg/mL)

Answer: A

12. A rat weighs 0.024 kg and requires 100 mg/kg of a 10 mg/mL drug solution. The volume required is:

A. 0.12 mL B. 0.24 mL C. 0.48 mL D. 2.40 mL

Answer: B

13. A rat weighs 0.022 kg and requires 100 mg/kg of a 10 mg/mL solution. The volume required is:

A. 0.11 mL B. 0.22 mL C. 0.33 mL D. 2.20 mL

Answer: B

14. A rat weighs 0.028 kg and requires 100 mg/kg of a 10 mg/mL solution. The volume required is:

A. 0.14 mL B. 0.28 mL C. 0.56 mL D. 2.80 mL

Answer: B

15. What instrument is used to administer the drug orally to rodents?

A. Intravenous catheter B. Oral feeding tube (gavage cannula) C. Tuberculin dropper D. Scalp vein needle

Answer: B

16. What timing device is explicitly listed in the Materials?

A. Sand timer B. Stopwatch C. Pendulum clock D. Metronome

Answer: B

17. The first route of administration tested in the Method is:

A. Oral B. Subcutaneous C. Intraperitoneal (i/p) D. Intramuscular

Answer: C

18. What physical maneuver is performed on the animal during observation?

A. Suspended by the tail B. Turned to lie on its back C. Placed in a swimming chamber D. Pinched on the earlobe

Answer: B

19. The specific behavioral endpoint recorded is the:

A. Time of onset of convulsions B. Time of loss of righting reflex (LORR) C. Time of complete cessation of respiration D. Time of pupil dilation

Answer: B

20. Loss of righting reflex (LORR) in laboratory rodents is a surrogate marker for:

A. Animal death B. Loss of consciousness / onset of surgical hypnotic anesthesia C. Peripheral nerve damage D. Dehydration

Answer: B

21. The four routes compared in Experiment 5 are:

A. Intravenous, Intrathecal, Oral, Sublingual B. Oral, Intraperitoneal, Intramuscular, Subcutaneous C. Topical, Inhalation, Rectal, Intradermal D. Buccal, Nasal, Intracardiac, Epidural

Answer: B

22. Which of the tested routes is classified as an enteral route?

A. Intraperitoneal B. Intramuscular C. Oral D. Subcutaneous

Answer: C

23. Which of the following is NOT one of the parenteral routes evaluated in this experiment?

A. Intraperitoneal B. Intravenous C. Intramuscular D. Subcutaneous

Answer: B

24. In the lab notebook, why was the intravenous (IV) route NOT used in mice?

A. Barbiturates cannot be injected intravenously B. Mice do not have clear veins that can easily be injected unlike larger rats C. Intravenous injection is instantly toxic D. Intravenous injection produces no pharmacological effect

Answer: B

25. The time between drug administration and the loss of righting reflex represents:

A. Duration of action B. Onset of action C. Elimination half-life D. Clearance rate

Answer: B

26. The time between the loss of righting reflex and recovery of the reflex represents:

A. Onset of action B. Duration of action C. Bioavailability D. Volume of distribution

Answer: B

27. Among the four tested routes, which typically produces the fastest onset of drug action?

A. Oral B. Subcutaneous C. Intraperitoneal (i/p) D. Intramuscular

Answer: C

28. The rapid onset of action following intraperitoneal injection is primarily due to:

A. Absence of capillaries in the abdomen B. Large absorptive surface area and rich mesenteric vascular supply C. Bypassing hepatic circulation completely D. Direct absorption into the cerebral ventricles

Answer: B

29. Among the four tested routes, which typically produces the slowest onset of action?

A. Intraperitoneal B. Intramuscular C. Oral D. Subcutaneous

Answer: C

30. The delay in drug onset following oral administration is primarily due to:

A. Rapid pulmonary excretion B. Gastric emptying time, dilution by gut contents, and mucosal barrier transit C. Immediate binding to skeletal muscle D. Complete destruction by salivary amylase

Answer: B

31. Drugs absorbed from the gastrointestinal tract and peritoneal cavity enter the portal circulation and undergo:

A. First-pass hepatic metabolism B. Renal tubular secretion C. Pulmonary filtration D. Complete biliary excretion unchanged

Answer: A

32. Between intramuscular (IM) and subcutaneous (SC) routes, which generally has faster onset?

A. Subcutaneous, because skin is thin B. Intramuscular, because skeletal muscle has higher vascular perfusion than subcutaneous fat C. Both have identical onset times in all species D. Subcutaneous, because adipose tissue accelerates absorption

Answer: B

33. How does the presence of food in the stomach affect oral drug onset?

A. Accelerates absorption 10-fold B. Delays onset by slowing gastric emptying rate or binding drug C. Prevents drug elimination permanently D. Has zero effect on absorption kinetics

Answer: B

34. How does high lipid solubility affect the onset of a central nervous system depressant?

A. Prevents passage across biological membranes B. Facilitates rapid crossing of the blood-brain barrier, accelerating onset C. Traps the drug permanently in the blood D. Accelerates renal filtration

Answer: B

35. Drug solutions generally act faster than solid dosage forms (tablets) because:

A. Tablets do not dissolve in water B. Solutions are already dissolved and do not require disintegration and dissolution C. Solutions bypass the liver completely D. Solutions contain zero active ingredients

Answer: B

36. In the experimental data (Page 44), which route showed the shortest average onset of action?

A. Oral B. Intraperitoneal (130.9 s) C. Intramuscular (860.5 s) D. Subcutaneous (1244.2 s)

Answer: B

37. In the experimental data (Page 44), which route showed the longest average onset of action?

A. Intraperitoneal B. Intramuscular C. Subcutaneous (1244.2 s) D. Oral

Answer: C

38. In the results table, what unexpected event was noted in group 7 during the experiment?

A. The animal gained weight rapidly B. The mouse had seizures and eventually died C. The animal became hyperactive and aggressive D. The drug turned bright red in the syringe

Answer: B

39. The expected general order of onset speed across the four routes from fastest to slowest is:

A. Oral > Subcutaneous > Intramuscular > Intraperitoneal B. Intraperitoneal > Intramuscular > Subcutaneous > Oral C. Subcutaneous > Oral > Intraperitoneal > Intramuscular D. Oral > Intraperitoneal > Subcutaneous > Intramuscular

Answer: B

40. The major conclusion of Experiment 5 is that:

A. The route of administration does not influence drug action B. The route of administration critically dictates the pharmacokinetic onset and duration of drug action C. All routes produce identical onset and duration if dose is constant D. Only oral administration is effective for barbiturates

Answer: B